

Project Review: Hybrid Energy SVD (Conv + FC1)

1. The Big Picture

Hey Sir, here is the review report for our Hybrid ESVD experiment. This time, we didn't just target the memory bottleneck (FC1); we went after the **Compute/Latency Bottleneck** by simultaneously compressing the Convolutional layer (Conv2). By using the Auto-Tuner to sweep for the perfect τ (tau) threshold across both layer types, we proved that we can drastically slash both physical file size and computational MACs (FLOPs) in a single pass.

2. Auto-Tuning: The Threshold Sweep

We instructed the auto-tuner to test thresholds down to 0.70, stopping if the raw accuracy dropped by more than 1.0%.

Threshold (τ)	Raw Accuracy	Accuracy Drop
0.99	98.89%	0.00%
0.95	98.92%	-0.03% (Increase!)
0.90	98.88%	0.01%
0.85	98.83%	0.06%
0.80 (Winner)	98.37%	0.52%
0.75 (Aborted)	97.79%	1.10% (Passed 1.0% limit)

The Verdict: The script dynamically halted at $\tau = 0.80$ because pushing to 0.75 crossed our 1.0% drop limit. This means we are retaining 80% of the mathematical energy across both the spatial and dense matrices. The raw accuracy took a very manageable 0.52% hit, which sets us up perfectly for a lossless healing phase.

3. Let's Do the Manual Math

To prove the hourglass SVD logic is working flawlessly on both architectures, let's reverse-engineer the parameter counts from the torchinfo logs.

A. The Dense Layer (FC1) Calculation

Original FC1 was 3136 inputs → 128 outputs (**401,536 parameters**). Our script selected an optimal rank of **k1 = 20** (out of a max possible 128).

- **Linear A (3136 → 20):** $3136 * 20 = 62,720$ weights (No bias)
- **Linear B (20 → 128):** $20 * 128 = 2,560$ weights + 128 biases = 2,688
- **Total Compressed FC1:** $62,720 + 2,688 = 65,408$ params

B. The Convolutional Layer (Conv2) Calculation

Original Conv2 was 32 input channels → 64 output channels with a 3x3 kernel (**18,496 parameters**). The script unfolded this 4D tensor and selected an optimal rank of **k2 = 16** (out of a max possible 64).

- **Conv A (Spatial Filter):** 32 in → 16 out, 3x3 kernel. $(16 * 32 * 3 * 3) = 4,608$ weights (No bias).
- **Conv B (Pointwise Mixer):** 16 in → 64 out, 1x1 kernel. $(64 * 16 * 1 * 1) = 1,024$ weights + 64 biases = 1,088.
- **Total Compressed Conv2:** $4,608 + 1,088 = 5,696$ params

C. The Net Savings

- **FC1 Params Deleted:** $401,536 - 65,408 = 336,128$
- **Conv2 Params Deleted:** $18,496 - 5,696 = 12,800$
- **Final Model Check:** Original 421,642 - $(336,128 + 12,800) = 72,714$ total parameters remaining. This matches the torchinfo console output perfectly!

4. Hardware Efficiency: Solving Both Bottlenecks

By compressing both layers, we achieved an incredible Pareto balance. Look at the computational operations (MACs):

Metric	Baseline	Hybrid ESVD Compressed	Total Reduction
Memory (Parameters)	421,642	72,714	82.75% Saved
Compute (MACs)	~4.28 Million	~1.43 Million	~66.5% Saved

5. Conclusion

This experiment proves the true power of Hybrid ESVD. When we only compressed FC1, our MACs barely moved (only ~8% saved). By targeting Conv2 as well, we mathematically deleted over **2.8 Million repetitive multiplication operations (MACs)** per image while maintaining a tiny 72k parameter footprint.

With a raw accuracy drop of only 0.52%, this model is primed to easily recover past the 99% accuracy mark during a fine-tuning phase, resulting in a lightning-fast, ultra-lightweight Edge AI network.